SMOOTH CENTRALLY SYMMETRIC CONVEX DOMAINS IN HIGH DIMENSIONS WITH INTERIOR HOT SPOTS

ARGUS OPEN RESEARCH PROJECT

Abstract. We document the counterexample of de Dios Pont (2024) to the hot-spots conjecture for centrally symmetric smooth convex domains in $\mathbb{R}^d$ for sufficiently high dimension d.

1. Main Refutation

Theorem 1.1 (de Dios Pont, 2024)**.** *For all sufficiently large dimensions d, there exists a smooth, bounded, centrally symmetric convex domain $\Omega \subset \mathbb{R}^d$ such that every second Neumann eigenfunction u of $-\Delta$ on Ω attains its strict global maximum and minimum in the interior of Ω rather than on the boundary $\partial\Omega$.*

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